CONDITION BASED ASSESSMENT
QUICK SCANNING REPORT
SBUM DIAGNOSTIC JOHOR
TNB BAHAGIAN PEMBAHAGIAN JOHOR

FUNCTIONAL LOCATION (ERMS)	:	JJBU/PCE/J03951
FUNCTIONAL LOCATION (SITE)	:	JJBU/PCE/J03951
SUBSTATION NAME (ERMS)	:	SK KG MAJU JAYA
SUBSTATION NAME (SITE)	:	SEK. KEB. KG. MAJU JAYA
COORDINATE	:	1.579787, 103.718929
SUBSTATION TYPE	:	PCE
AREA	:	JOHOR BAHRU
STATE	:	JOHOR
PO	:	42234207
DATE	:	14 JUL 2026

SCANNED BY		
THERMOGRAPHY	:	MUHAMMAD OMAR RIDZUAN (2013MY20N010)

EQUIPMENT/TOOLS		
THERMOGRAPHY	:	FLIR T640 (SN 55912229) (CALIBRATION DUE DATE: 26 DEC 2026)

PREPARED BY		
NAME OF CONTRACTOR	:	**EET ENGINEERING SERVICES SDN BHD**
ADDRESS	:	NO.38, JALAN SS19/1D,
	:	47500, SUBANG JAYA, SELANGOR.

SUBSTATION SIGNBOARD



EXECUTIVE SUMMARY

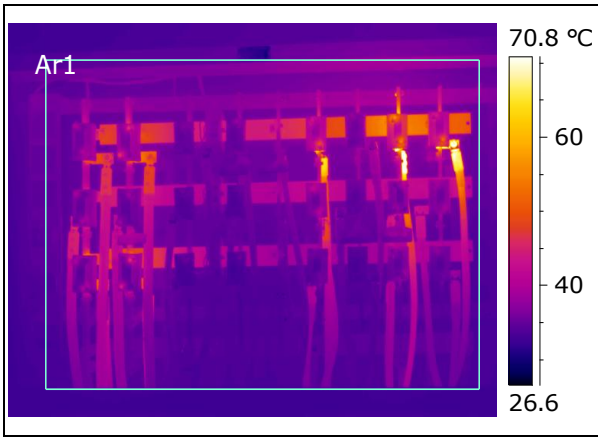
NO.	EQUIPMENT	DEFECT AREA	IR (Abs.T)	IR (ΔT)	DEFECT
1.	FP TX1 – INCOMING F1	INCOMING LINK CONNECTION/ RED PHASE	66.7 °C	14.8 °C	**REPAIR IMMEDIATELY**
2.	FP TX1 – OUTGOING F4	OUTGOING FUSE CONNECTION/ RED PHASE	106.2 °C	61.2 °C	**REPAIR IMMEDIATELY**
3.	FP TX1 – OUTGOING F6	OUTGOING FUSE CONNECTION/ RED PHASE	149.2 °C	66.6 °C	**REPAIR IMMEDIATELY**
4.	FP TX1 – OUTGOING F7	OUTGOING FUSE CONNECTION/ RED PHASE	219.2 °C	167.6 °C	**REPAIR IMMEDIATELY**

VISUAL DEFECT SUMMARY

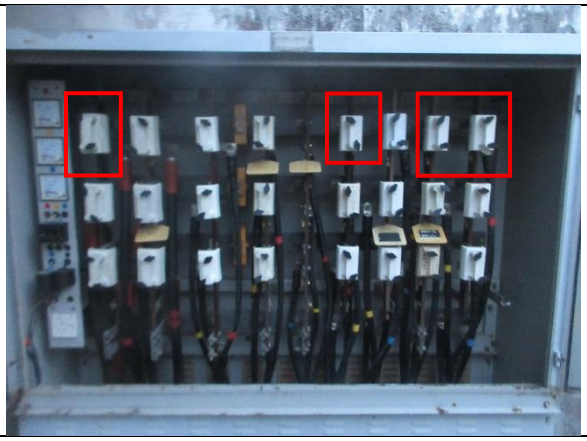
NO.	EQUIPMENT	DEFECT DESCRIPTION	ELECTRICAL	NON-ELECTRICAL
1	SUBSTATION	NO PADLOCK MAIN GATE		X
2	LIGHTING	MISSING	X	
3	FIRE EXTINGUISHER	MISSING		X

	Substation	SK KG MAJU JAYA			Date: 14/07/2026
	Equipment	FP TX1	Manufacturer	TOPRANK	Time: 07:40 PM
	Model	J-SLOTTED	Rating	1600A	Humidity: 63%
	Area	OVERVIEW		Feeder No.	-
Feeder Name	-				

IR IMAGE



VISUAL IMAGE



PARAMETER	VALUE
Spot Temp.1 (Sp1)	-
Spot Temp. 2 (Sp2)	-
Average Temp. (Ar1)	37.1 °C
Ambient Temp.	32.0 °C
Delta T (Δ T)	-
Load (A)	-
Severity	-

PARAMETER	VALUE
Serial No.	652
Heater (A)	-
CB Position	-
Busbar Position	-
Cable Type	-

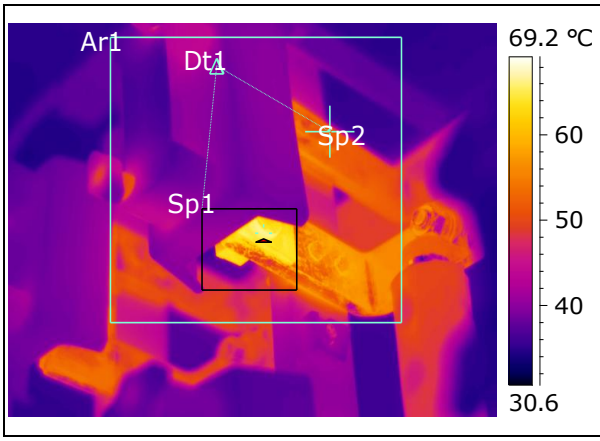
Analysis & Recommendations:

Analysis: Please refer the following page for details defect.

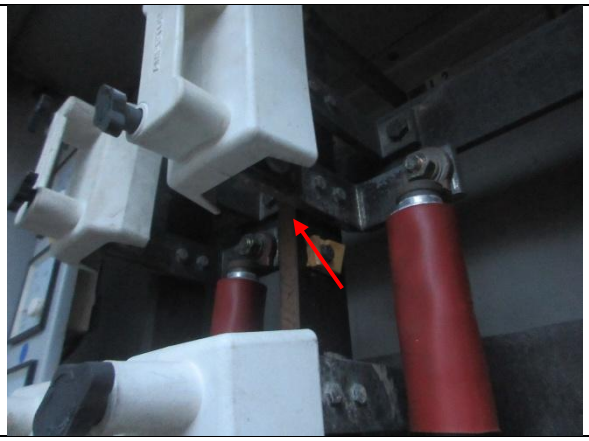
Recommendation: N/A

	Substation	SK KG MAJU JAYA			Date: 14/07/2026
	Equipment	FP TX1	Manufacturer	TOPRANK	Time: 07:40 PM
	Model	J-SLOTTED	Rating	1600A	Humidity: 63%
	Area	LINK CONNECTION		Feeder No.	INCOMING F1
Feeder Name	INCOMING F1 – RED PHASE				

IR IMAGE



VISUAL IMAGE



PARAMETER	VALUE
Spot Temp.1 (Sp1)	66.7 °C
Spot Temp. 2 (Sp2)	51.9 °C
Average Temp. (Ar1)	44.2 °C
Ambient Temp.	32.0 °C
Delta T (Δ T)	14.8 °C
Load (A)	-
Severity	REPAIR IMMEDIATELY

PARAMETER	VALUE
Serial No.	652
Heater (A)	-
CB Position	-
Busbar Position	-
Cable Type	XLPE

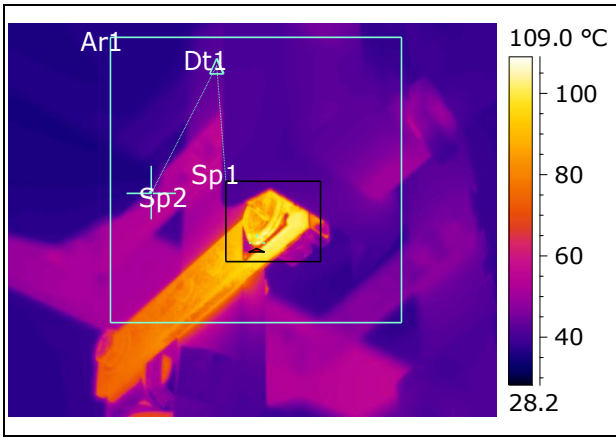
Analysis & Recommendations:

Analysis: Thermal image above indicates hot spot detected at **INCOMING LINK CONNECTION.** The anomaly is due to loosen or bad contact at the connection point.

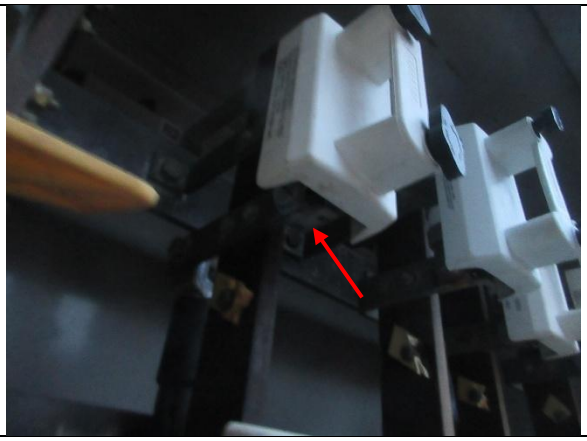
Recommendation: To inspect, clean the contact surface and re-tighten the connection point. Replace defective parts if necessary.

	Substation	SK KG MAJU JAYA			Date: 14/07/2026
	Equipment	FP TX1	Manufacturer	TOPRANK	Time: 07:40 PM
	Model	J-SLOTTED	Rating	1600A	Humidity: 63%
	Area	FUSE CONNECTION		Feeder No.	OUTGOING F4
Feeder Name	OUTGOING F4 – RED PHASE				

IR IMAGE



VISUAL IMAGE



PARAMETER	VALUE
Spot Temp.1 (Sp1)	106.2 °C
Spot Temp. 2 (Sp2)	45.0 °C
Average Temp. (Ar1)	48.8 °C
Ambient Temp.	32.0 °C
Delta T (Δ T)	61.2 °C
Load (A)	-
Severity	REPAIR IMMEDIATELY

PARAMETER	VALUE
Serial No.	652
Heater (A)	-
CB Position	-
Busbar Position	-
Cable Type	XLPE

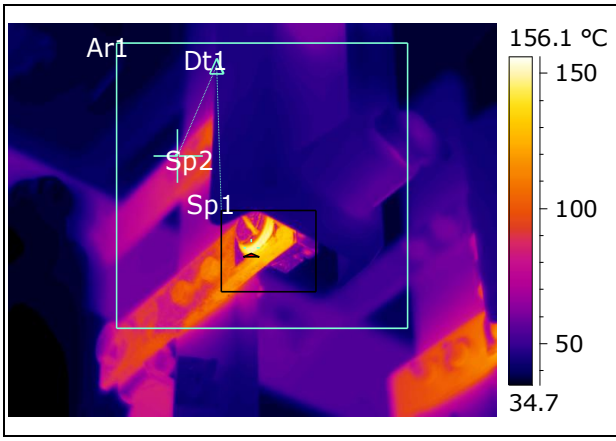
Analysis & Recommendations:

Analysis: Thermal image above indicates hot spot detected at **OUTGOING FUSE CONNECTION.** The anomaly is due to loosen or bad contact at the connection point.

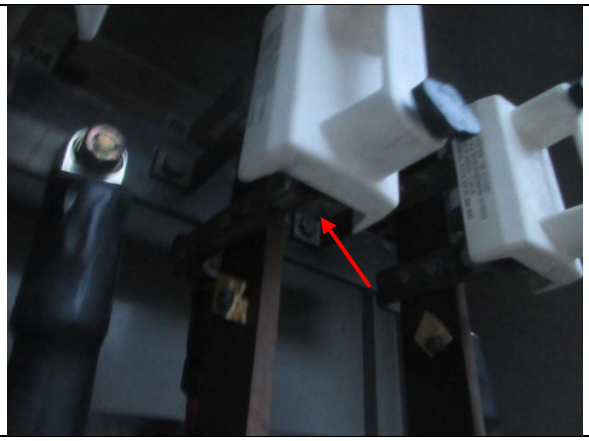
Recommendation: To inspect, clean the contact surface and re-tighten the connection point. Replace defective parts if necessary.

	Substation	SK KG MAJU JAYA			Date: 14/07/2026
	Equipment	FP TX1	Manufacturer	TOPRANK	Time: 07:40 PM
	Model	J-SLOTTED	Rating	1600A	Humidity: 63%
	Area	FUSE CONNECTION		Feeder No.	OUTGOING F6
Feeder Name	OUTGOING F6 – RED PHASE				

IR IMAGE



VISUAL IMAGE



PARAMETER	VALUE
Spot Temp.1 (Sp1)	149.2 °C
Spot Temp. 2 (Sp2)	82.6 °C
Average Temp. (Ar1)	56.2 °C
Ambient Temp.	32.0 °C
Delta T (Δ T)	66.6 °C
Load (A)	-
Severity	REPAIR IMMEDIATELY

PARAMETER	VALUE
Serial No.	652
Heater (A)	-
CB Position	-
Busbar Position	-
Cable Type	XLPE

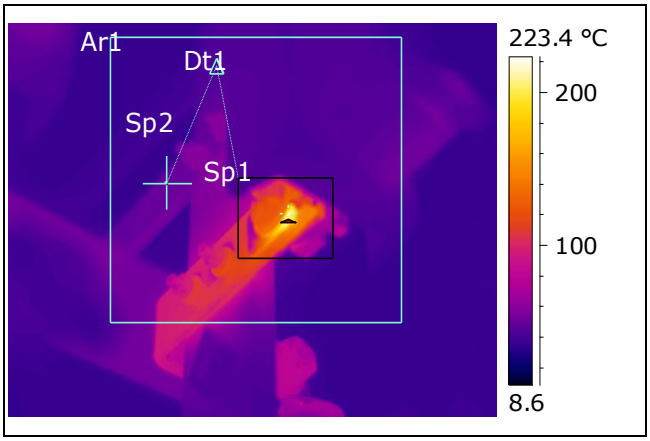
Analysis & Recommendations:

Analysis: Thermal image above indicates hot spot detected at **OUTGOING FUSE CONNECTION.** The anomaly is due to loosen or bad contact at the connection point.

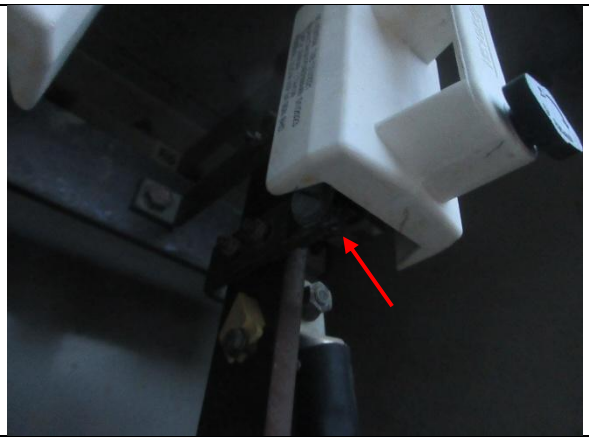
Recommendation: To inspect, clean the contact surface and re-tighten the connection point. Replace defective parts if necessary.

	Substation	SK KG MAJU JAYA			Date: 14/07/2026
	Equipment	FP TX1	Manufacturer	TOPRANK	Time: 07:40 PM
	Model	J-SLOTTED	Rating	1600A	Humidity: 63%
	Area	FUSE CONNECTION		Feeder No.	OUTGOING F7
Feeder Name	OUTGOING F7 – RED PHASE				

IR IMAGE



VISUAL IMAGE



PARAMETER	VALUE
Spot Temp.1 (Sp1)	219.2 °C
Spot Temp. 2 (Sp2)	*51.6 °C
Average Temp. (Ar1)	*53.5 °C
Ambient Temp.	32.0 °C
Delta T (Δ T)	*167.6 °C
Load (A)	-
Severity	REPAIR IMMEDIATELY

PARAMETER	VALUE
Serial No.	652
Heater (A)	-
CB Position	-
Busbar Position	-
Cable Type	XLPE

Analysis & Recommendations:

Analysis: Thermal image above indicates hot spot detected at **OUTGOING FUSE CONNECTION.** The anomaly is due to loosen or bad contact at the connection point.

Recommendation: To inspect, clean the contact surface and re-tighten the connection point. Replace defective parts if necessary.

SUBSTATION CONDITION

SUBSTATION OVERVIEW



SIGNBOARD



SWITCHGEAR



SWITCHGEAR NAMEPLATE



TRANSFORMER 1



TRANSFORMER 1 NAMEPLATE



SUBSTATION CONDITION

TRANSFORMER 2
N/A

TRANSFORMER 2 NAMEPLATE
N/A

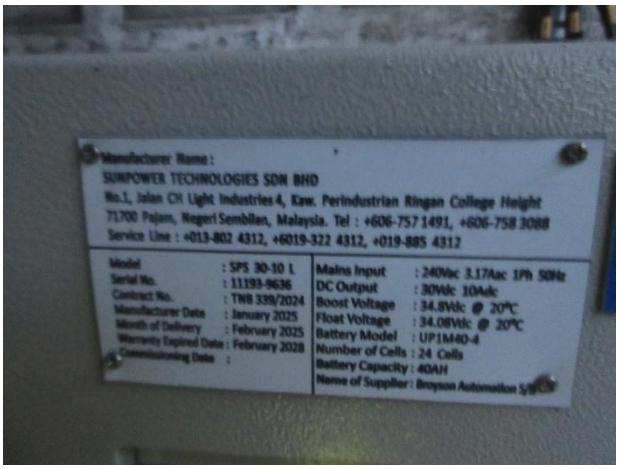
FEEDER PILLAR TX1


FEEDER PILLAR TX1 NAMEPLATE


FEEDER PILLAR TX2
N/A

FEEDER PILLAR TX2 NAMEPLATE
N/A

SUBSTATION CONDITION

EFI	SF6 INDICATOR
	
BATTERY	**BATTERY NAMEPLATE**
	
FIRE EXTINGUISHER	**FIRE EXTINGUISHER NAMEPLATE**
N/A	**N/A**

SUBSTATION CONDITION

SWITCHGEAR PANEL NAME	LAST PREVENTIVE MAINTENANCE
	

VISUAL DEFECT

SUBSTATION



NO PADLOCK MAIN GATE

LIGHTING



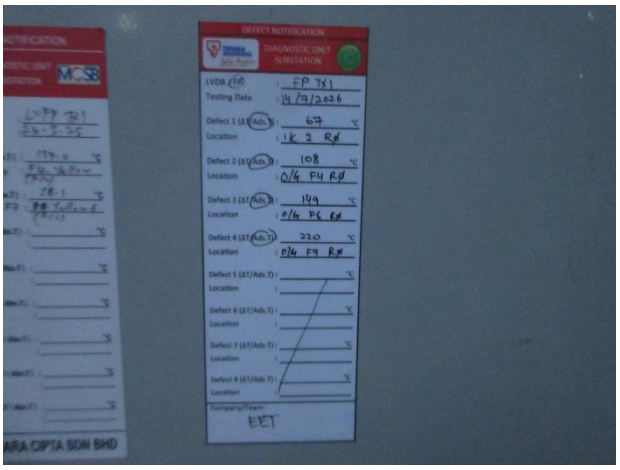
MISSING

FIRE EXTINGUISHER



MISSING

NORMAL/DEFECT STICKER

NORMAL STICKER	DEFECT STICKER
	

CBM CONDITION: 80 DEFECT

DISTRIBUTION SUBSTATION (PCE) CBA TEST SHEET

STATE / AREA:	JOHOR / JOHOR BAHRU	**DATE**	14/07/26		**FUNCTIONAL LOCATION:**
PE NAME:	SK KG MAJU JAYA	**TIME IN / OUT**	1940 /	2030	JJBU/PCE/J03951
V.I REF. NO	80	**TEV BACK GROUND / HUMIDITY**	5 dB /	63 %	**BACKGROUND TEMP : 32.0 °C**

SWITCHGEAR

DETAIL								TEMPERATURE							ULTRASOUND			TEV			
NO.	FDR. NO	FEEDER NAME	STATUS (TRIP / CLOSE)	LOAD (A)	CABLE TYPE/SIZE	HEATER (A)	SERIAL NO (ONLY BREAKER & OLU)	COMP	Tmin (°C)	Tmax (°C)	ΔT	Avg (°C)	FILE	REMARKS	dB	FILE	TYPE CHAR.	dB	PULSE/ CYCLE	TYPE CHAR.	REMARK
1	JJB96243	ULAM RAJA	CLOSE		XLPE			CABLE	32.6	32.8	0.2	32.7	227		0			5	0		
								BREAKER													
								TOP PANEL													
								PT													
2	JJB96244	TX 1000 KVA	CLOSE		XLPE			CABLE	32.4	32.6	0.2	32.5	228		0			5	0		
								BREAKER													
								TOP PANEL													
								PT													
3	JJB96245	SETIA B/PARK 2 NO 15	CLOSE		XLPE			CABLE	32.6	32.6	0.0	32.6	229		0			5	0		
								BREAKER													
								TOP PANEL													
								PT													
4								CABLE													
								BREAKER													
								TOP PANEL													
								PT													
OVERVIEW								CABLE				33.2	222								
								BREAKER													
								TOP PANEL				32.2	226								
								PT													

LOCAL TRANSFORMER

DETAILS				THERMOGRAHY					ULTRASOUND		DETAILS			THERMOGRAHY						ULTRASOUND		
TX ID	SERIAL NO	CABLE TYPE/SIZE	REMARK	COMP.	Tmin(°C)	Tmax(°C)	ΔT	Avg (°C)	FILE	dB	TYPE CHAR.	TX ID	SERIAL NO	CABLE TYPE/SIZE	COMP.	Tmin(°C)	Tmax(°C)	ΔT	Avg (°C)	FILE	dB	FILE	TYPE CHAR.	REMARKS
TX1	EWT/91670	XLPE		HT CABLE	33.3	33.5	0.2	33.4	233	0		TX3			HT CABLE									
				HT BUSHING	38.3	38.6	0.3	38.5	232	0					HT BUSHING									
		XLPE		LV CABLE	36.0	36.3	0.3	36.2	235						LV CABLE									
				LV BUSHING	36.0	36.2	0.2	36.1	234						LV BUSHING									
				BODY	39.8	44.5	4.7	42.2	230						BODY									
TX2				HT CABLE								TX4			HT CABLE									
				HT BUSHING											HT BUSHING									
				LV CABLE											LV CABLE									
				LV BUSHING											LV BUSHING									
				BODY											BODY									

LV DISTRIBUTION BOARD/FEEDER PILLAR/OTHERS

1	LVDB / FP 1	INC. 1	INC. 2		INC. 3		F1	F2	F3	F4	F5	F6	F7	F8	F9	F10					
	CABLE TYPE / SIZE	XLPE	XLPE				XLPE	XLPE		XLPE	XLPE	XLPE	XLPE								
2	LVDB / FP 2	INC. 1	INC. 2		INC. 3		F1	F2	F3	F4	F5	F6	F7	F8	F9	F10					
	CABLE TYPE / SIZE																				

NO	LVDB / FP	DEFECTIVE FEEDER NAME	Tmin(°C)	Tmax(°C)	ΔT	FILE	Avg (°C)	REMARK
	INC 1							
FPTX1	INC 2	RED PHASE 67	51.9	66.7	14.8	237	59.3	INCOMING LINK CONNECTION
	F1							
	F2							
	F3							
	F4	RED PHASE 108	45.0	106.2	61.2	238	75.6	OUTGOING FUSE CONNECTION
	F5							
	F6	RED PHASE 149	82.6	149.2	66.6	239	115.9	OUTGOING FUSE CONNECTION
	F7	RED PHASE 220	51.6	219.2	167.6	240	135.4	OUTGOING FUSE CONNECTION
	F8							
	Battery Bank					241		BC1: SUNPOWER M:SPS 30-10L SN: 11193-9636
	Black Box 1							
	Black Box 2							
	Lightning Arrester							
	Dropout Fuse							
	Auto Recloser							
	LBS							

FP TX1	J-SLOTTED
Manufacturer :	TOPRANK
Serial No. :	652
Rating :	1600A
LVDB TX2	
Manufacturer :	
Serial No. :	
Rating :	

Tested by (Name & Signature) :

EET ENGINEERING SERVICES SDN BHD
(534556-W)
No. 38-2, 2nd Floor, Jalan SS 19/1D
47500 Subang Jaya, Selangor.
Tel: 603-5636 6201 Fax: 603-5636 7205

TNB Supervisor/Engineer (Name & Signature) :

NOR AZMI BIN ABDUL RAHMAN
(N/P 10013600)
Pen. Juruteknik T/Kanan
Diagnostik (Substation - South)
Distribution Network Department, TNB

Ver. : 1-2019

vi fe miss, light miss, n plock mg

TYPE PCE 80

VISUAL INSPECTION CHECKLIST FOR 11kV SUBSTATION

TENAGA NASIONAL BERHAD (200866-W)

DATE:	14/07/26												
P/E Name (ERMS):	SK KG MAJU JAYA							F/L NO (ERMS):	JJBU/PCE/J03951				
P/E Name (SITE):	SEK. KEB. KG. MAJU JAYA							F/L NO (SITE):	JJBU/PCE/J03951				
GPS Coordinate:	1.579787, 103.718929												
PCE Type:	INDOOR			OUTDOOR	☑	COMPACT		ATTACH BUILDING		PAT		OTHER	
PCE Area	NORMAL	☑		INDUSTRIAL		COASTAL							
Switcgear Type 1:	RMU OIL			RMU SF6		MRMU	☑	VCB		OCB		OTHER	
Manufacturer/Model:	EPE			CFC 2S+1F									
Switchgear Manufactured Date:	2024					Switchgear Rating :		12kV				RMU SF6 / OIL S/N :	23050047400
Switcgear Type 2:	RMU OIL			RMU SF6		MRMU		VCB		OCB			
Manufacturer/Modal:													
Switchgear Manufactured Date:						Switchgear Rating :						RMU SF6 / OIL S/N :	
No. of Transformer:	1												
Tx 1:	Type	H/S		KVA	1000kVA	Construction Yr:	2024	Manufacturer:	EWT			S/N :	EWT/91670
Tx 2:	Type			KVA		Construction Yr:		Manufacturer:				S/N :	
Tx 3:	Type			KVA		Construction Yr:		Manufacturer:				S/N :	
Tx 4:	Type			KVA		Construction Yr:		Manufacturer:				S/N :	

KEJANGGALAN

1. Signboard:	Good	☑	Not Good		Remarks	
2. Earth connection:	Good	☑	Not Good		Remarks	
3. EFI Functioning:	Good	☑	Not Good		Remarks	
4. Heater Functioning:	Good		Not Good		Remarks	N/A
5. Labelling @Switchgear:	Good	☑	Not Good		Remarks	
6. Labelling @ LVDB:	Good	☑	Not Good		Remarks	
7. Gas Pressure:	Good	☑	Not Good		Remarks	
8. Tx Guard:	YES	☑	NO		Remarks	
9. Tx Oil Level (conservator tank):	Good		Not Good		Remarks	N/A
10. Tx Oil Leak:	YES		NO	☑	Remarks	
11. Tx Bushing:	Good	☑	Not Good		Remarks	
12. Top Demand Indicator (@LVDB):	Good	☑	Not Good		Remarks	
13. Link/fuse Holder:	Good	☑	Not Good		Remarks	
14. LV Termination:	Good	☑	Not Good		Remarks	
15. Rubber Mat:	YES	☑	NO		Remarks	
16. Trenching:	Good	☑	Not Good		Remarks	
17. Battery Voltage:	Good	☑	Not Good		Remarks	
18. Lighting:	Good		Not Good	☑	Remarks	
19. Roof Condition:	Good	☑	Not Good		Remarks	
20. Fire Fighting	Valid		Expired		Remarks	Expiry Date: - No. of missing portable : 1
21. Fence/door condition:	Good	☑	Not Good		Remarks	
22. Cleanliness/Bushes:	YES		NO	☑	Remarks	
23. Vandalisme	YES		NO	☑	Remarks	
24. Other Findings:						

Tested by (Name & Signature) :

EET ENGINEERING SERVICES SDN BHD
(534556-W)
No. 38-2, 2nd Floor Jalan SS 19/1D
47500 Subang Jaya, Selangor.
Tel: 603-5636 6205 Fax: 603-5636 7205

Date :

TNB Supervisor / Engineer (Name & Signature) :

NOR AZMI BIN ABDUL RAHMAN
(N/P 10073600)
Pen. Juruteknik T/Kanan
Diagnostik (Substation - South)
Distribution Network Department, TNB

Date :

Ver. : 1-2019